BABA BOOEY

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

Alice and Bob take turns playing a game, with Alice going first. Before the first move, there is an array of non-negative integers $a_1, a_2, \cdots, a_n$ of length n, along with a positive integer k. In each move, the current player chooses two integers x, y such that $0 \leq x \leq k$, $1 \leq y \leq n$, and $x \leq a_y$. Then, k is updated to x, and a_y is updated to $a_y - x$. A player who chooses $x = 0$ loses immediately. Who has the winning strategy?

Input

There are multiple test cases. The first line contains an integer t ($1 \leq t \leq 10^5$) — the number of test cases.

For each test case:

The first line contains two integers n and k ($1 \leq n \leq 10^6$, $1 \leq k \leq 10^9$).

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($0 \leq a_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases in a single input file does not exceed 2×10^6.

Output

For each test case, output a single string on a new line.

If the first player (Alice) has a winning strategy for the game described, output "`Alice`"; otherwise, output "`Bob`".

Example

standard input	standard output
3 2 2 1 2 4 5 15 15 8 9 11 2 4 2 9 12 0 15 9 14 5 6 12	Alice Alice Bob

Note

In the first test case, the only winning first moves for Alice are $(x, y) = (1, 1)$ and $(x, y) = (1, 2)$.